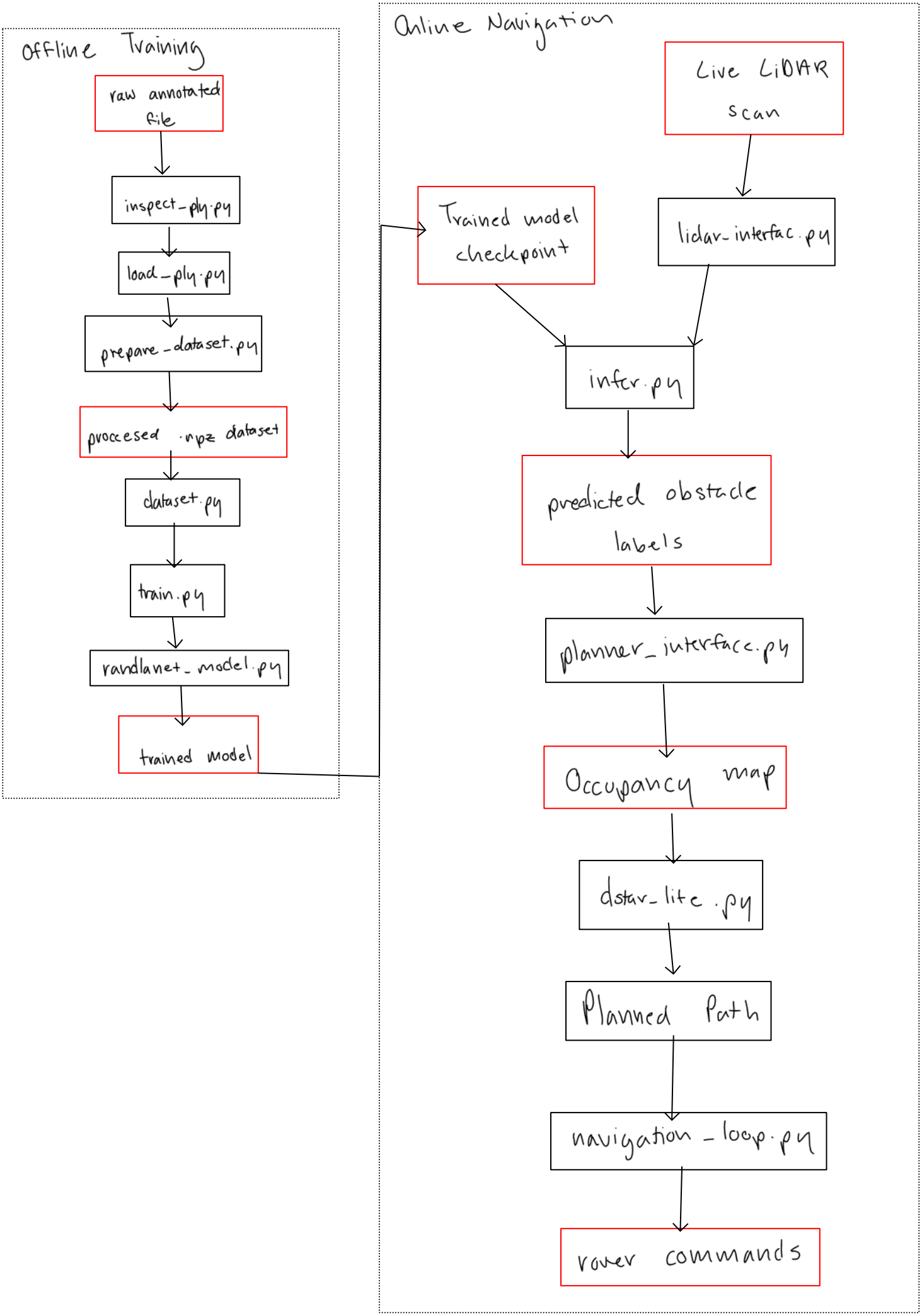
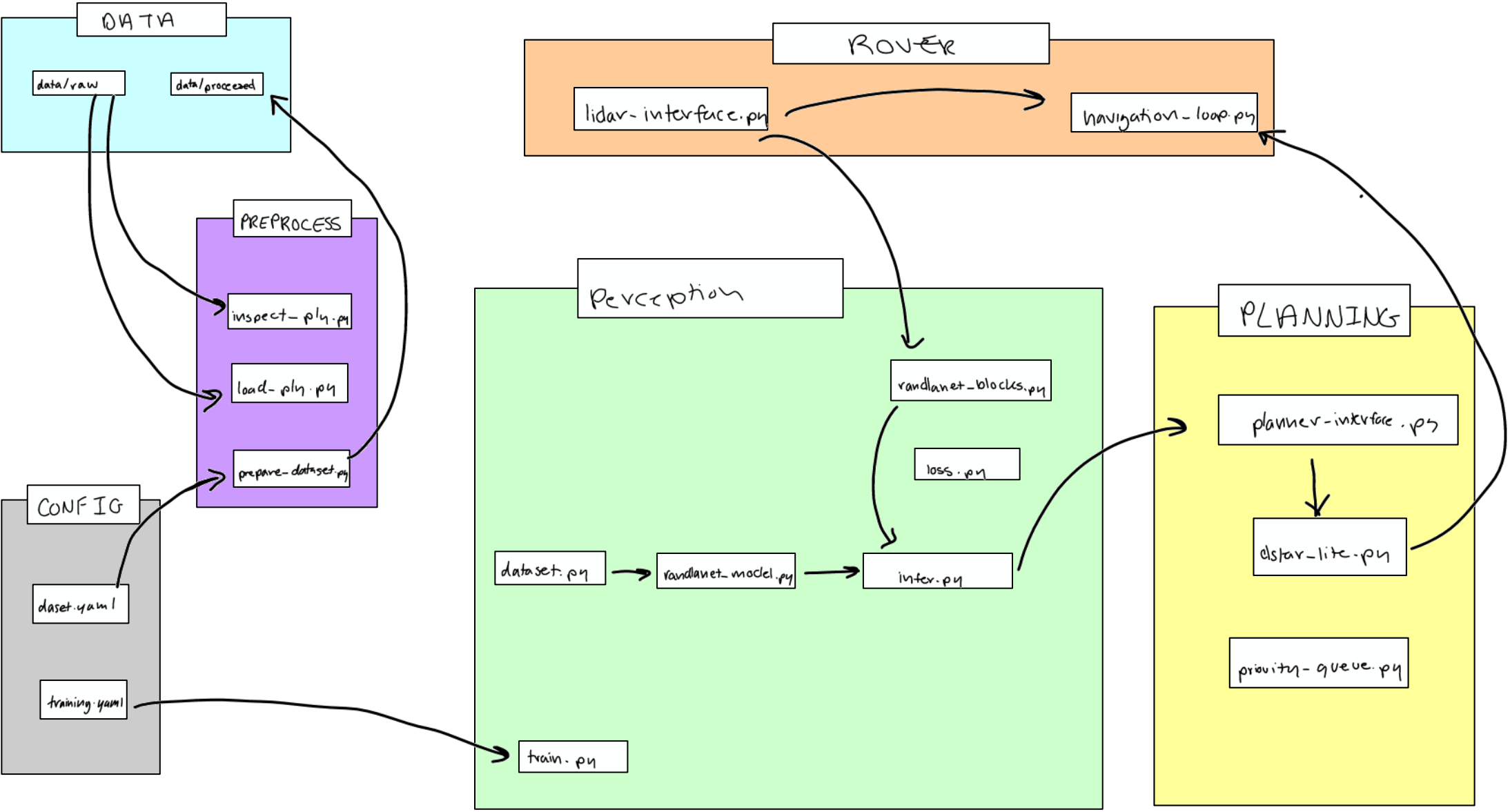
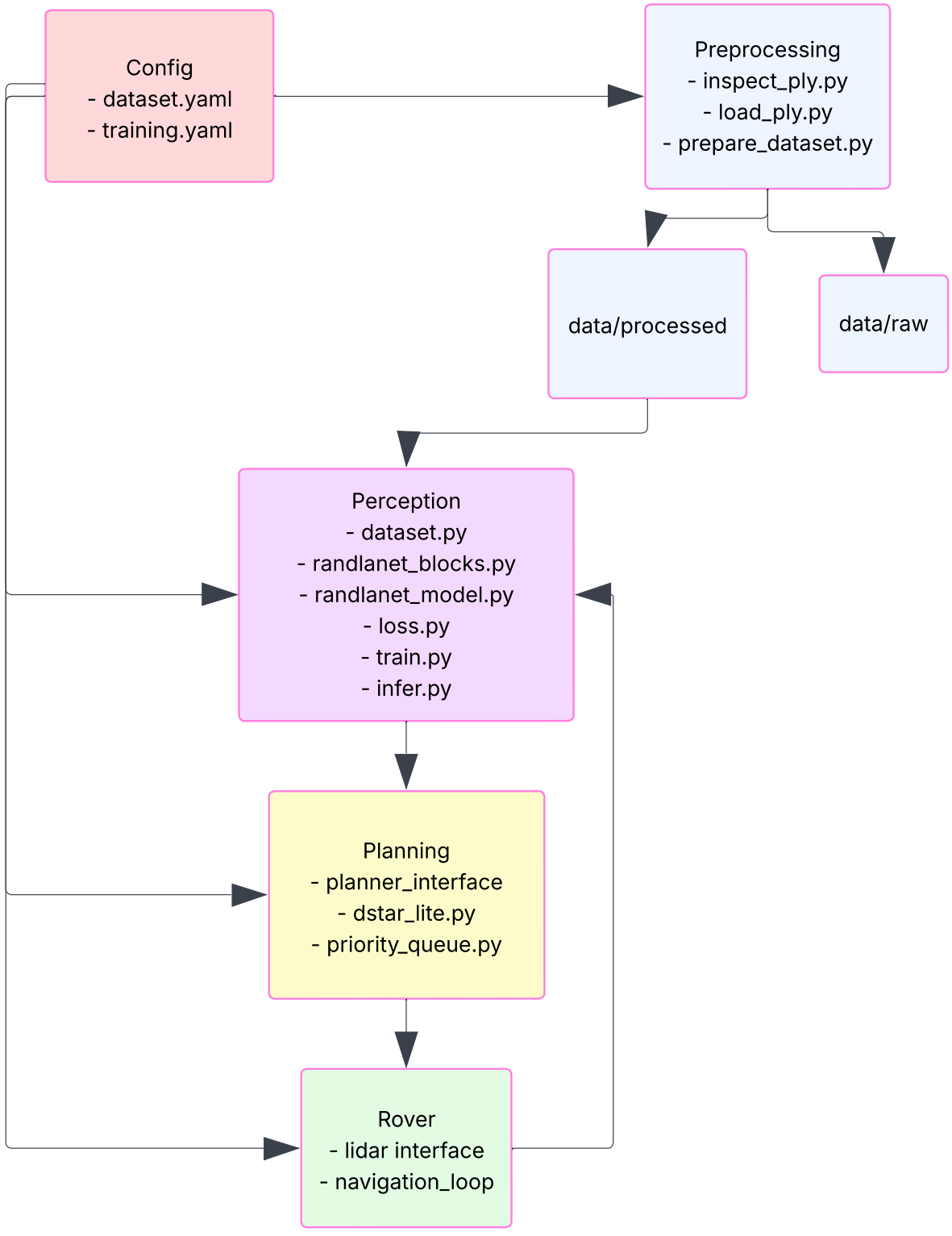
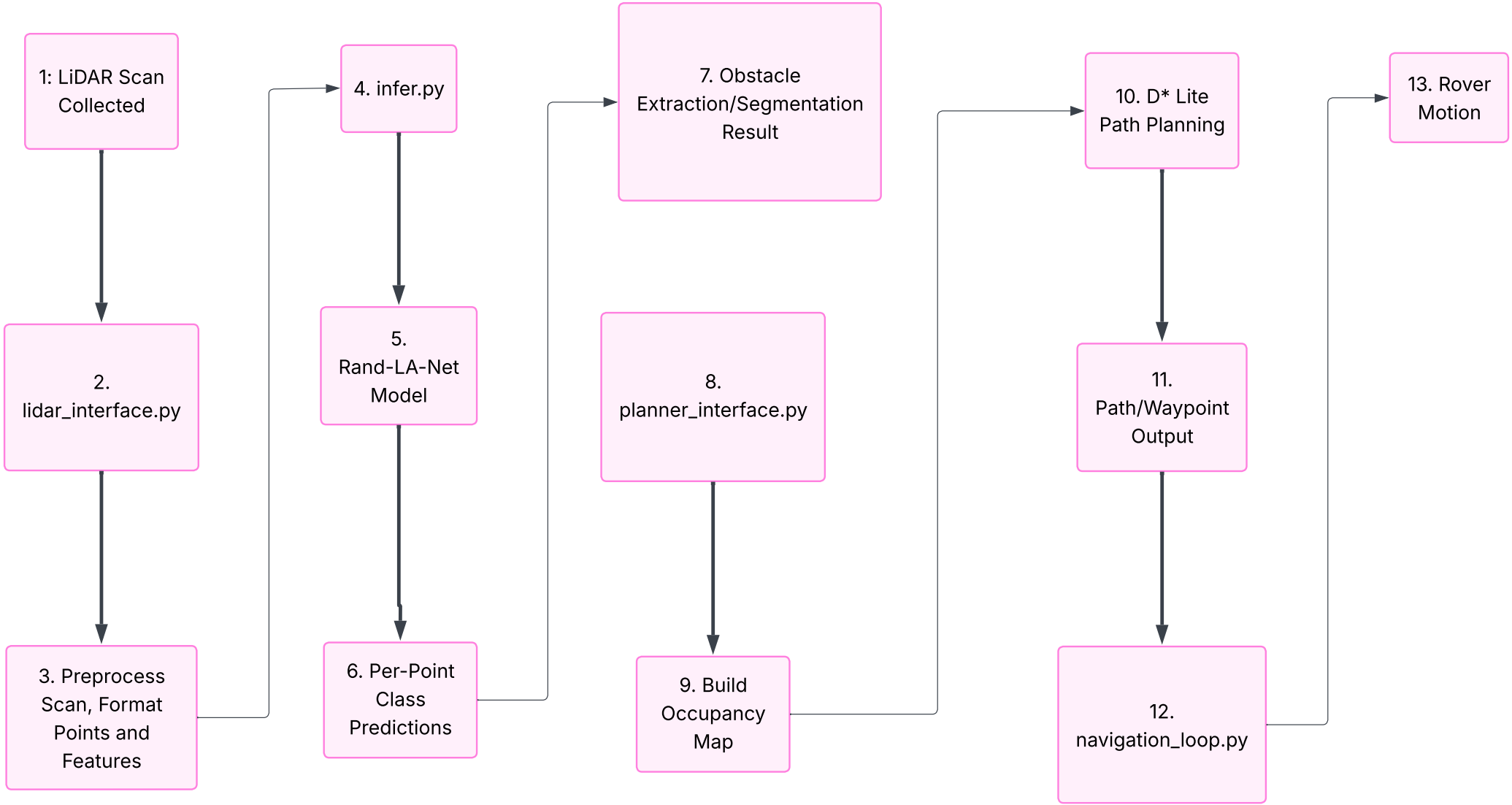

Offline Training
raw annotated file
inspect_ply.py
load_ply.py
prepare_dataset.py
proccesed .npz dataset
dataset.py
train.py
randlanet_model.py
trained model
Online Navigation
Live LiDAR scan
Trained model checkpoint
lidar_interfac.py
infer.py
predicted obstacle labels
planner_interface.py
Occupancy map
dstar_lite.py
Planned Path
navigation_loop.py
rover commands


DATA
data/raw
data/proceaed
PREPROCESS
inspect_ply.py
load_ply.py
prepare_dataset.py
CONFIG
daset.yaml
training.yaml
ROVER
lidar-interface.py
navigation_loop.py
Perception
randlanet_blocks.py
loss.py
dataset.py
randlanet_model.py
infer.py
train.py
PLANNING
planner-interface.py
dstar_lite.py
priority-queue.py

Software System Architecture Diagram




1: LiDAR Scan Collected
2. lidar_interface.py
3. Preprocess Scan, Format Points and Features
4. infer.py
5. Rand-LA-Net Model
6. Per-Point Class Predictions
7. Obstacle Extraction/Segmentation Result
8. planner_interface.py
9. Build Occupancy Map
10. D* Lite Path Planning
11. Path/Waypoint Output
12. navigation_loop.py
13. Rover Motion